MILITARY SPECIFICATION

CUTTING MACHINES, OXYGEN FLAME AND PLASMA ARC, AUTOMATIC, STATIONARY, NUMERICALLY CONTROLLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, stationary, automatic, flame and plasma arc cutting machines for cutting various shapes from steel plate or other materials. The plasma arc system used with the cutting machines is covered by MIL-C-80090.

1.2 Classification. This specification covers cutting machines of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - For cutting with an oxygen-fuel gas torch, having straight line and contour cutting speeds with a range of not greater than 2 inches per minute (IPM) to not less than 40 IPM.

Type II - For cutting with a plasma arc torch, or oxygen-fuel gas torch, having straight line cutting speed with a range of not greater than 50 IPM to not less than 240 IPM and contour cutting speed of not greater than 2 IPM to not less than 120 IPM.

Size 1 - 48 in. width x 96 in. length cutting capacity.
Size 2 - 60 in. width x 120 in. length cutting capacity.
Size 3 - 72 in. width x 144 in. length cutting capacity.
Size 4 - 96 in. width x 192 in. length cutting capacity.
Size 5 - 120 in. width x 240 in. length cutting capacity.
Size 6 - 144 in. width x 288 in. length cutting capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

MIL-C-80090 - Cutting System, Metal, Plasma Arc.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine.

RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Electrical and Construction Standards for Numerical Machine Control.

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-447 - Operational Command and Data Format for Numerically Controlled Machines.

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W. Washington, DC 20006.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of cutting material in accordance with the requirements herein. The machine shall be all purpose capable of performing intricate shape and straight line cutting operations using a numerically controlled (NC) coordinated drive system for the combination of transverse and longitudinal movement. The machine shall be capable of multiple torch operation. The design shall be such as to provide convenient, efficient, and safe operation with minimum maintenance. All structural components shall be designed to withstand operating conditions at maximum capacity while maintaining the accuracies specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

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3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Fuel. Unless otherwise specified, the machine shall be equipped for use with acetylene fuel gas. When specified (see 6.2.1), the cutting machine shall be equipped for use with the fuel gas specified.

3.2.10 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.11 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of

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these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial standard practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components.

3.4.1 Carriage assembly. The carriage assembly shall be constructed of metal with sufficient strength and rigidity to insure performance as specified herein. The rail and wheel contact surfaces shall have a finish that will assure vibration-free operation and performance as specified herein. Coordinated transverse and longitudinal movements shall be provided by individual drive motors, with provisions to lock the drive in transverse or longitudinal position for making straight cuts, parallel cuts, and cuts at 90 degrees to the main rails. Driving surfaces of the carriage shall not be exposed to the cutting operation. Movements of the carriage shall be positive, with backlash between drive and driven members limited to the extent necessary to assure cutting operations within the tolerances specified in 3.6. A suitable air blast or wiper shall be incorporated, if necessary, to keep rail surfaces clean. The gas and electrical distributing system shall be integral with the carriage or cross beam, and shall be mounted thereon. Unless otherwise specified, the machine shall be furnished for right hand operation. Right hand operation shall be with the torches to the right and the controls to the left of the operator when facing the control station. When specified (see 6.2.1), the machine shall be furnished for left hand operation. Left hand operation shall be with the torches to the left and the controls to the right of the operator when facing the control station.

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3.4.2 Carriage rails. Steel carriage rails shall be of the manufacturer's standard design and shall be of a type which readily permits extension by adding sections. When specified (see 6.2.1), additional sections shall be furnished. Travel stops or limit switches which prevent carriage overtravel shall be provided.

3.4.3 Coordinate drive. The coordinate drive shall provide directional movements for the machine carriage. The carriage shall be powered by servo motors which are coordinated electronically through a directional control unit actuated by the machine control.

3.4.4 Cutting torch, oxygen-fuel gas. The oxygen-fuel gas cutting torch shall be of the valved, 3 hose machine type, with a cylindrical barrel and rack, and shall be designed for use with oxygen and a specified gas. The end of the torch, housing the fuel and oxygen valves, shall be manufactured of bronze or brass. The torch head shall be constructed to receive cutting tips of various sizes. Unless otherwise specified (see 6.2.1), four oxygen cutting torches shall be furnished with the type I machines and three with the type II machines.

3.4.5 Torch ignition. A remote controlled automatic ignitor shall be provided for each torch.

3.4.6 Cutting system plasma arc. When specified (see 6.2.1), the type II machine shall be equipped with a plasma arc cutting system. The plasma arc system shall conform to MIL-C-80090.

3.4.7 Motorized torch holder. Unless otherwise specified (see 6.2.1), the machine shall be equipped with motorized torch holders as follows: type I machines shall be equipped with four holder assemblies designed for mounting cylindrical oxygen cutting torches as specified in 3.4.4; type II machines shall be equipped with three holder assemblies designed for mounting cylindrical oxygen cutting torches, and one holder assembly designed for mounting a plasma arc cutting torch conforming to MIL-C-80090. The construction of the holder assemblies shall provide for manually positioning the assemblies at any point along the carriage, and for removal of the assemblies from the carriage. Provisions shall be made for tilting the torches forward for high speed ripping and tilting sideways for beveling. The vertical travel of the torches shall be sufficient for cutting material up to 8 inches thick. The torch holder suspension system shall be of sufficient strength and rigidity to prevent binding or distortion in any manner and to assure operation within the tolerances specified herein. When specified (see 6.2.1), special torch holder suspension systems shall be furnished as required.

3.4.8 Numerical control system. The numerical control shall be a fully automatic, soft wired, integrated circuit system. The system shall provide

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control of all cutting torches and movement of the machine "X" and "Y" axes. The axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of the two axes of machine movement. Control features shall include buffer storage, part program storage, program edit capability, and control diagnostics. The system shall direct machine functions from part program data stored in memory or from manual data input (MDI). The control shall initiate a halt or an error signal should a fault condition occur in the control. All necessary executive program routines shall be stored in nonvolatile memory. The system shall be constructed in accordance with EIA Standard RS-281. The control shall be capable of operating in ambient temperatures ranging from 40 to 120 degrees Fahrenheit and relative humidity from 5 to 95 percent. A line voltage variation of ±10 percent from nominal shall not adversely affect the control system function. The control shall have power protection to provide an orderly controlled shutdown when a power failure occurs. Control resolution shall be 0.001 inch for each linear axis.

3.4.8.1 Data input.

3.4.8.1.1 Punched tape data input. The data input media shall be one inch, eight channel, punched tape. The data input format and codes shall comply with EIA Standards RS-244, RS-274, RS-358, and RS-447. The control shall be equipped with a photoelectric tape reader capable of holding, feeding, and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard 5-1/4 inch diameter feed and take-up spools. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the display screen to indicate a fault condition.

3.4.8.1.2 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of all characters necessary for control of all programmable machine functions. The MDI shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.8.2 Data display. Unless otherwise specified (see 6.2.1), either a cathode ray tube (CRT) or a universal alphanumeric data display unit shall be provided. When only one type of data display is acceptable, the particular type required shall be specified (see 6.2.1). The display unit shall be capable of presenting at least 250 characters simultaneously. The display

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unit shall be capable of being used in a variable scanning mode to interrogate all registers sequentially or held to display only a particular register as the content changes. Either unit shall display, as a minimum, the following data:

a. Sequence number.
b. Position of each axis.
c. Control and system error messages.

3.4.8.3 Absolute/incremental input. The control shall have absolute and incremental input capability.

3.4.8.4 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 character codes.

3.4.8.5 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.8.6 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be of either solid state or bubble type memory. Unless otherwise specified (see 6.2.1), the control shall be capable of storing in memory the content of not less than 100 feet of postprocessed part program tape data.

3.4.8.7 Part program edit. A program edit capability shall be provided to allow deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input mode. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.8.8 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.8.9 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.8.10 Mirror image. The machine shall be capable of producing either right or left-hand parts from the same tape data through the inversion of the "X" and "Y" axes. Inversion of each axis shall be by keyboard or switch entry. Reverse settings shall be displayed on the data display.

3.4.8.11 Preparatory function (G-codes). The preparatory function codes of the control shall be in accordance with or comply with the intent of EIA Standard RS-274.

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3.4.8.12 Miscellaneous functions (M-codes). The miscellaneous function codes of the control shall be in accordance with or comply with the intent of EIA Standard RS-274.

3.4.8.13 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed dated.

3.4.8.14 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.8.15 Kerf compensation. The control shall allow kerf compensation up to 0.99 inches.

3.4.8.16 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than ±99.999 inches.

3.4.8.17 Floating zero. The control shall allow the zero reference set point of each axis to be established at any point within the work envelope.

3.4.8.18 Axis jog. The control shall have manual controls for accomplishing both single and continuous low and high speed jog movements.

3.4.8.19 Program tryout. The control shall have a program tape tryout feature to permit new part programs to be read through the control without cutting torches being turned on.

3.4.8.20 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or in the form of punched tape to provide for diagnostic checking of the control unit. The diagnostic software shall test, exercise, and display failures to board level.

3.4.8.21 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory to prevent loss of memory data for a period of not less than 70 hours when the primary power source fails.

3.4.9 Gas control. The oxygen and fuel gas regulators and their associated gauges shall be conveniently located for ease of reference and adjustment. The equipment shall include regulation for preheat fuel, preheat oxygen, and the cutting oxygen. An individual solenoid valve to control the cutting oxygen shall be furnished for each torch. A system for controlling preheat and fast piercing starts shall be furnished. Flow of gases shall be automatically controlled by solenoid valves which shall immediately close when the master control switch is turned to the off position. Regulation and manifolding shall be of sufficient capacity to accommodate the specified

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number of torches. The manifold shall have capped outlets to permit future installation of additional torches in accordance with requirements listed in table I. All other associated gas control equipment, including hoses and fittings necessary for internal machine connection and operation, shall be included. All components shall be mounted and protected to withstand the operating conditions encountered when the machine is operated at maximum rated load and speed. When hose lengths exceed 7 feet, suitable means shall be provided for relieving the weight and preventing entanglement and chafing of the hoses to each torch.

3.4.10 Operator control panel. The system shall have an operator control panel which provides at least the following control functions: main power "ON and OFF" switch, selector to control cutting oxygen mode, control of fuel gas to selected torches, and control of motorized torch suspension.

3.4.11 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Unless otherwise specified (see 6.2.1), electrical input power shall be 115 volts, single phase, 60 Hz, AC.

3.4.12 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors not mounted in enclosed compartments shall be of the totally enclosed type. Motor horsepower and speed shall be adequate to meet maximum operating conditions.

3.4.13 Solid state components. Solid state design shall be used throughout for electronic components.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machine shall be not less than those specified in table I for the machine size specified.

TABLE I. Sizes and capacities.

Capacities	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6
Width of cut, inches (transverse)	48	60	72	96	120	144
Length of cut, inches (longitudinal)	96	120	144	192	240	288
Diameter-circle cut, inches (min.-max.)	3-48	3-60	3-72	3-96	3-120	3-144
Manifolds (number of)	4	6	6	6	8	8

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3.6 Performance. The machine and all components shall function properly to meet the requirements specified herein. Accuracy of torch travel with respect to program path of the NC control shall be + 1/64 inch. The machine shall be capable of negotiating 90 degree turns with not more than a 1/8 inch radius at speeds up to 40 IPM. Machines equipped with 4 oxygen cutting torches shall be capable of cutting SAE 1020 or other low carbon steel 8 inches thick at a rate of not less than 3 IPM; 6 inches thick at a rate of not less than 5 IPM; and 1/4 inch thick at a rate of not less than 28 IPM. Straight line and contour cutting speed ranges for the type I and type II machines shall be as specified (see 1.2).

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the machine shall be furnished with all standard equipment furnished by the manufacturer.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking of instruction plates, indicating plates, and nameplates. All words on instruction plates, indicating plates, and nameplates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard cutting machine and a suffix identified in the manufacturer's permanent records. The

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captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to total time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1, 4.6.1.1, and the inspection in 4.7. Failure of the item to pass the examination, test, and inspection shall be cause for rejection.

4.5 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The machine shall be connected to the electric power source, the oxygen supply, and a specified fuel gas supply. Gas pressures shall be set in accordance with the supplier's recommendations. All components of the gas system shall be checked for leaks. All torches shall be ignited and adjusted to obtain a neutral flame cone pattern. Ignition and flame cycle of the torches shall be repeated 10 times per minute for a minimum of 3 minutes. During the test the torches shall maintain satisfactory continuous operation without adjustment, blowback, or backfire when the oxygen pressure is increased or decreased 6 PSI gauge pressure from its preselected setting. Any evidence of faulty or erratic operation shall be cause for rejection.

4.6.1.1 Numerical Control test. Functions of the machine and the control unit shall be checked in accordance with the manufacturer's standard procedure. The test shall include operating the machine in each of its modes of operation to determine proper response of the machine to control commands. The machine and control unit shall be operated in the test mode using a standard check tape or tapes to determine proper response to tape commands. Upon completion of the test the machine shall be cycled continuously for a period of two hours without a malfunction. Should a failure occur, the cycle shall be restarted after the correction and run for a full four hour period.

4.6.2 Depth of cut test. A sample of SAE 1020 or other low carbon steel 6 inches thick, 10 inches wide, and long enough to accommodate three or four torches as applicable, shall be cut. The time required for the 10 inch cut shall be not greater than two minutes. The cut edge of the kerf shall be square and the slag on the bottom of the cut shall be easily removable.

4.6.3 Speed of cut test. A sample of SAE 1020 or other low carbon steel 1/4 inch thick, 28 inches wide and long enough to accommodate the four torches shall be cut. The time required for the 28 inch cut shall be not greater than one minute. The cut edge of the kerf shall be square and the slag on the bottom of the cut shall be easily removable.

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4.6.4 Pattern test. The machine shall be programmed to run 90 degree angles, acute angles, and circles. For the purpose of this test, one torch shall be fitted with a tracing pen in lieu of the torch tip, and a sheet of drafting paper of the appropriate size shall be placed in the work area. The machine shall be operated and a complete tracing shall be made in the clockwise direction; then a retracing shall be made in the counterclockwise direction. This test shall be repeated no less than six times. Accuracy of torch travel with respect to programmed path shall be within ±1/64 inch accuracy.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The machine described in this specification will be used for cutting straight lines, circles, arcs, and various shapes from metals.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify torch fuel gas (see 3.2.9). If natural gas, specify line pressure.

h. Painting, if different (see 3.3.5).

i. Left hand operation, if required (see 3.4.1).

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j. Additional carriage rail sections, if required (see 3.4.2).

k. Number of oxygen-fuel gas cutting torches required, if different (see 3.4.4).

l. Plasma arc cutting system, if required (see 3.4.6).

m. Number of motorized torch holders, if different (see 3.4.7).

n. Type of torch holder suspension system, if different (see 3.4.7).

o. Description of tape reader, if different (see 3.4.8.1.1).

p. Data display, if a specific type is required (see 3.4.8.2).

q. Specific type of memory, if different (see 3.4.8.6).

r. Part program storage, if different (see 3.4.8.6).

s. Length of memory protection, if different (see 3.4.8.21).

t. Electrical power requirement, if different (see 3.4.11).

u. Size and capacity, if different (see 3.5).

v. Additional performance features, if required (see 3.6.1).

w. Standard equipment, if different (see 3.7).

x. Optional equipment, if required (see 3.8).

y. Repair parts, if required (see 3.9).

z. Lubrication chart or plate, if different (see 3.10.1).

aa. Nameplate, if different (see 3.10.2).

bb. Hour meter, if required (see 3.11).

cc. First article inspection, if different (see 4.3).

dd. Quality conformance inspection, if different (see 4.4).

ee. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level (see 5.1).

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6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1 (f).

Custodian:
Air Force - 99

Review activities:
Air Force -84
DLA - GS

User Activities:
Navy - MC, SH

Preparing activity:
DLA - IP

(Project 3433-0105)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-80265	Cutting, Machine, Oxygen Flame and Plasma Arc, Automatic, Stationary, Numerically Controlled

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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